2022511

IET, DAVV, Indore

BE 2^{nd} Year Information Technology sec (A and B)

3ITRC4-Digital Electronics

TIME: 3Hours MAX MARKS:60

NOTE: Solve any two parts from each question. Each questions carry equal marks. Make suitable assumptions whenever required. Q.5 has internal choice.

Q.1 (a) Convert the following numbers into equivalent number system. 6

(i) $(100101.011)_2$ ----------- $(\ )_8$

(ii) $(25A6.1E)_{16}$ ------------ $(\ )_{10}$

(iii) $(425.25)_{10}$ ----------- $(\ )_2$

(iv) Find the 10's and 9's complement of (543)10

Q.1 (b) (i) Convert Gray number 1101 to its BCD equivalent. 6

(ii) Perform subtraction (+83) and (-27) using 2's complement.

(iii) write NATIONAL SERVICE SCHEME in ASCII code.

Q.1 (C) (i) Prove OR- AND configuration is equivalent to NOR-NOR configuration. 4

(ii)Design a logic circuit to convert 4-bit BCD to Gray code. 2

Q.2 (a) (i) Minimize the following expression using Quine McClusky Technique 4

$F(A, B, C, D) = \sum(1,2,5,7,9,15) + d(0,3,11)$

(ii) What is K-map? Where it is used? 2

Q.2 (b) Design 4-bit magnitude comparator circuit and explain its working in detail. 6

Q.2 (C) Define Multiplexer with the help of functional diagram and Design 1 × 8 Demultiplexer and explain its working. 6

Q.3 (a) (i)What is the advantage offered by an edge triggered RS flip flop over a clocked on RS flip-flop? 2

(ii) Design positive edge triggered D flip flop and explain it with the help of truth table and characteristics table. 4

Q.3 (b) Explain step by step procedure designing in synchronous counter. Design 3-bit synchronous down counter using JK flip-flop. 6

Q.3 (C) Design 4 bit asynchronous up-down counter and differentiate between synchronous and asynchronous sequential circuit. 3+3

Q.4 (a) Design parallel in serial out shift register and explain its working with the help of timing diagram. 6

Q.4 (b) Design 4-bit universal shift register and explain its working with input bit 1100 for each case. 6

Q.4 (C) The content of a 4-bit shift register is initially 1101. The register is shifted six times to the right, with the serial input being 101101.what is the content of the register after each shift? 6

Q.5 (a) Write short note on any one. 6

(a) Successive Approximations A/D converter

(b) Binary weighted register Digital to analog converter

Q.5 (b) In R-2R DAC, find the full-scale output voltage if $R_f = 1.5k\Omega$ and $R_i = 2k\Omega$ and also find the output voltage when the input is 11101. Assume $V_{ref} = 6V$ also find resolution and FSR. 6

—x— x —x—